

XGBoost: From First Principles

XGBoost is an optimized gradient boosting algorithm designed for speed, regularization, and strong performance on tabular data.

Why XGBoost Is Needed

Gradient Boosting is powerful but can be slow and prone to overfitting. XGBoost improves it with regularization, efficient tree building, and strong engineering choices.

Core Idea

XGBoost builds trees sequentially. Each new tree improves the current model by reducing the loss.

Objective Function

XGBoost optimizes prediction loss plus a regularization penalty:

$$\textit{Objective} = \textit{Loss} + \textit{Regularization}$$

The regularization term discourages overly complex trees.

Second-Order Approximation

XGBoost uses both first-order gradients and second-order gradients, which helps it make more informed updates.

Important Controls

Important ideas include learning rate, number of trees, tree depth, subsampling, column sampling, and regularization.

When to Use It

Use XGBoost for structured/tabular data when strong predictive performance matters. Be careful with tuning and overfitting.